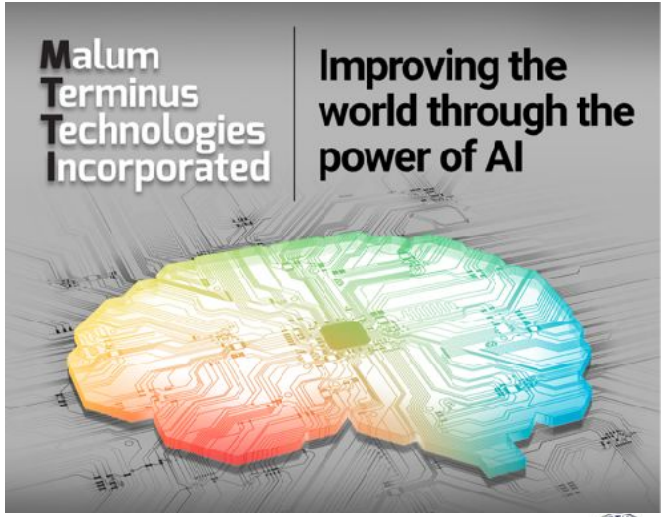
Intelligent Readiness

Personnel Readiness & Human-Centric Artificial Intelligence (AI)



March 2021

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Agenda

- Readiness, retention, injury
- Personnel Readiness and AI
- Current application: Musculoskeletal Injury (MSK-I)
- Future applications / discussion

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Human Assets: Crucial, complex

Recruiting → **Training & Development** → **Readiness & Retention** → **Retirement & Re-entry**

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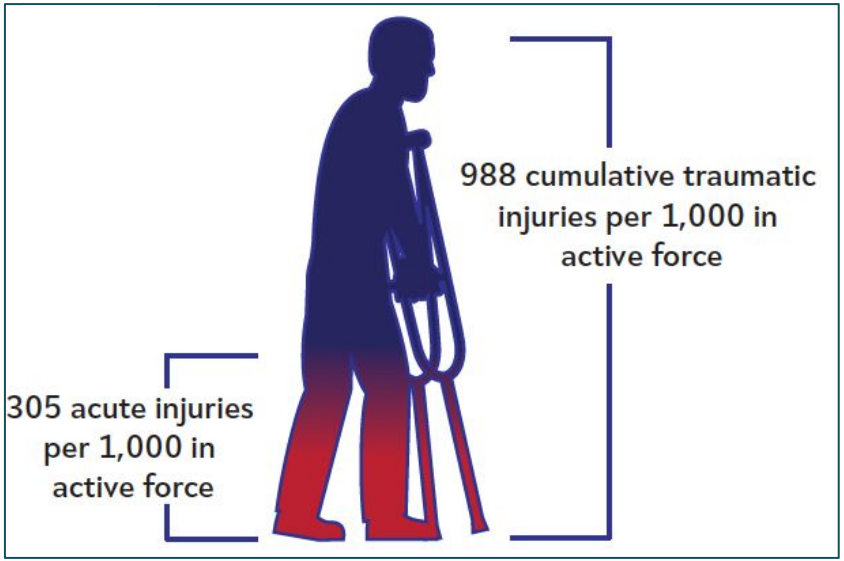
Injury: A massive, complex problem

BUSINESS INSIDER

"...the biggest challenge to readiness is injuries incurred during training"

October 17, 2019

#1 = Musculoskeletal



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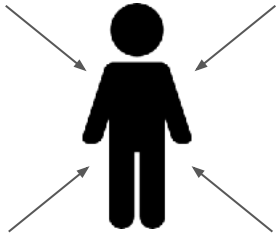
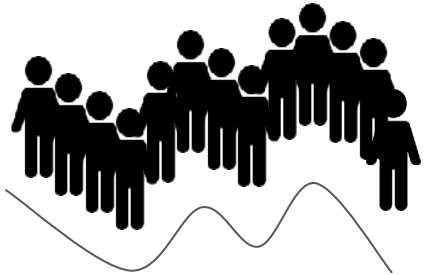
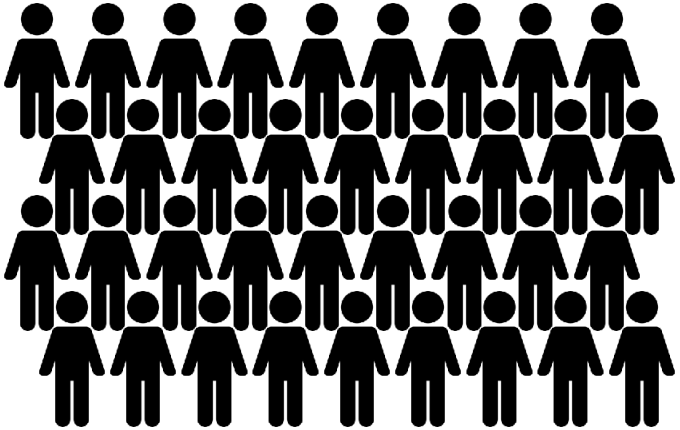
<u>Human System</u>: Most complex known



Unlimited Number of Variables



Constant Time Impacts & Changes



Everyone is Unique

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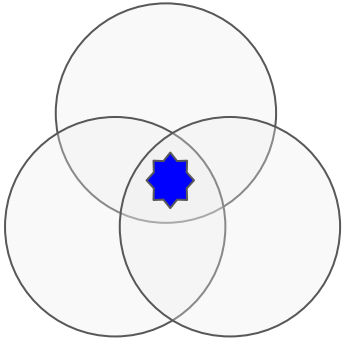
Breakthrough: Mitigation via Intervention

A paradigm shift

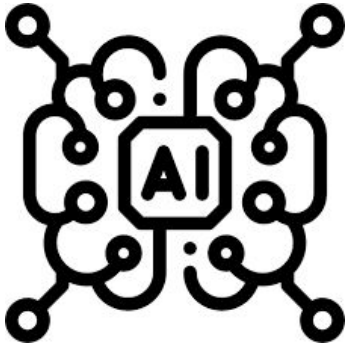
Human Performance,
Physiology & Kinesiology



Data Science &
Mathematics

Technology &
Machine Learning

Novel AI Technology



- Disparate data
- Innate relationships
- Dynamic prediction
- What-if simulation
- 1200X faster

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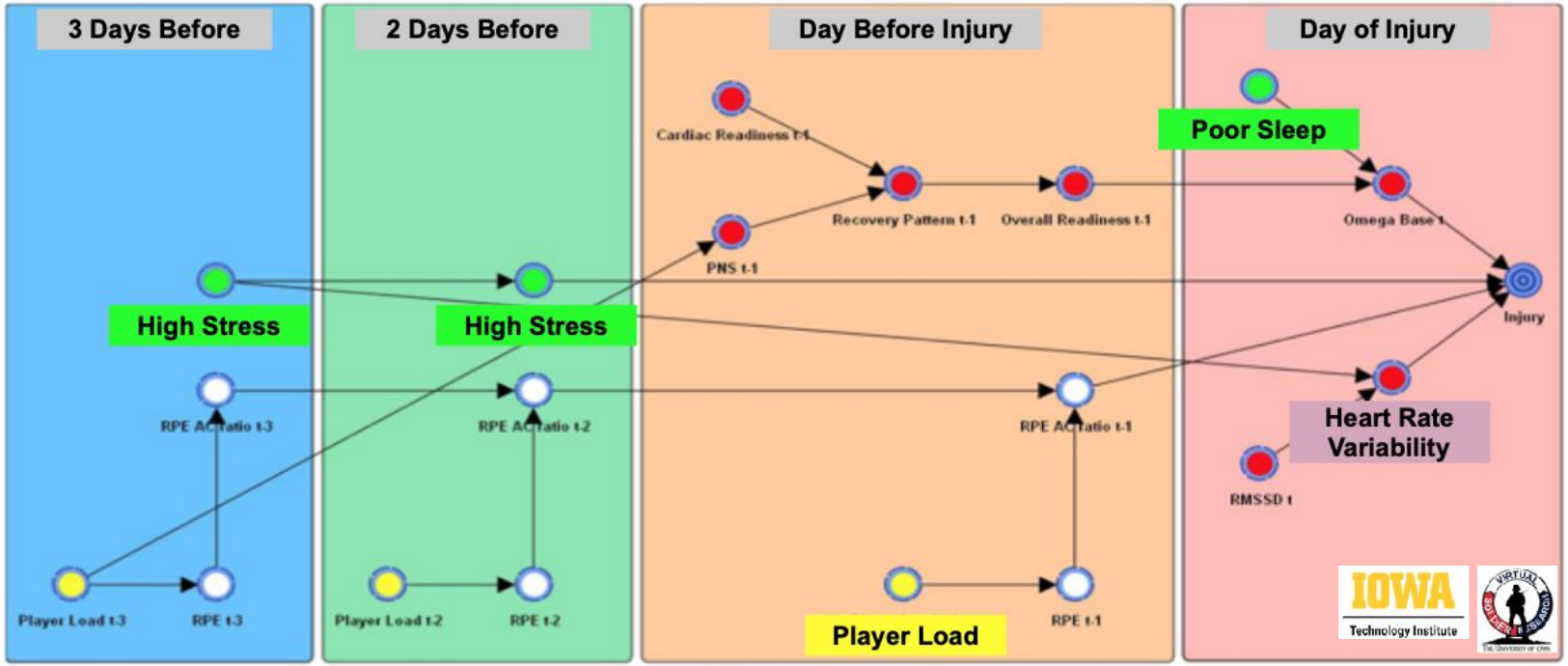
ONR Code 30 R&D: Daily metrics



It's not just about player / operator load; Injuries occur due to a variety of complex factors - some occurring days prior to the injury

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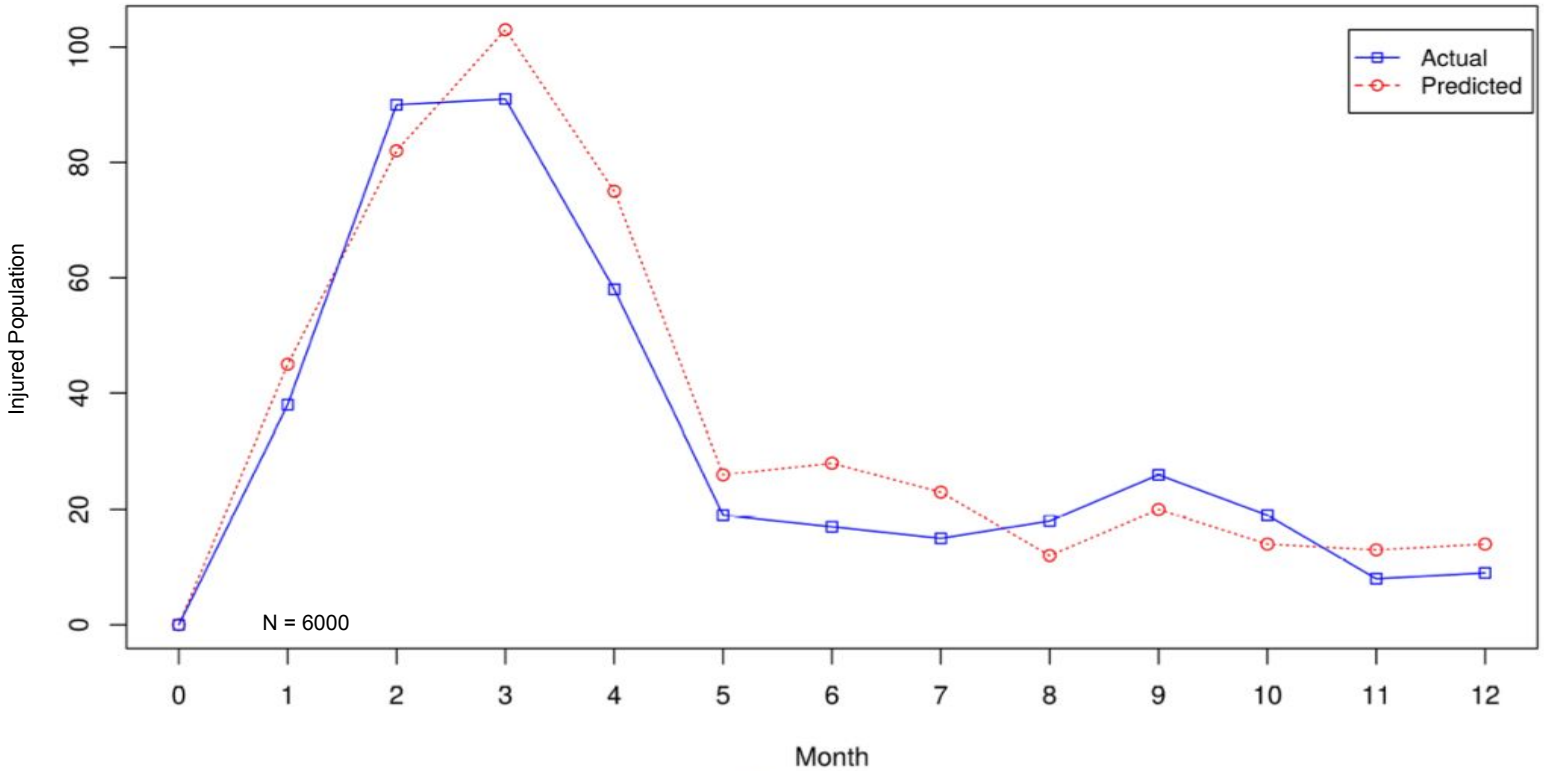
Warfighter: Annual / Point-in-time metrics



High-degree of "time-to-event" accuracy despite limited, low-fidelity data

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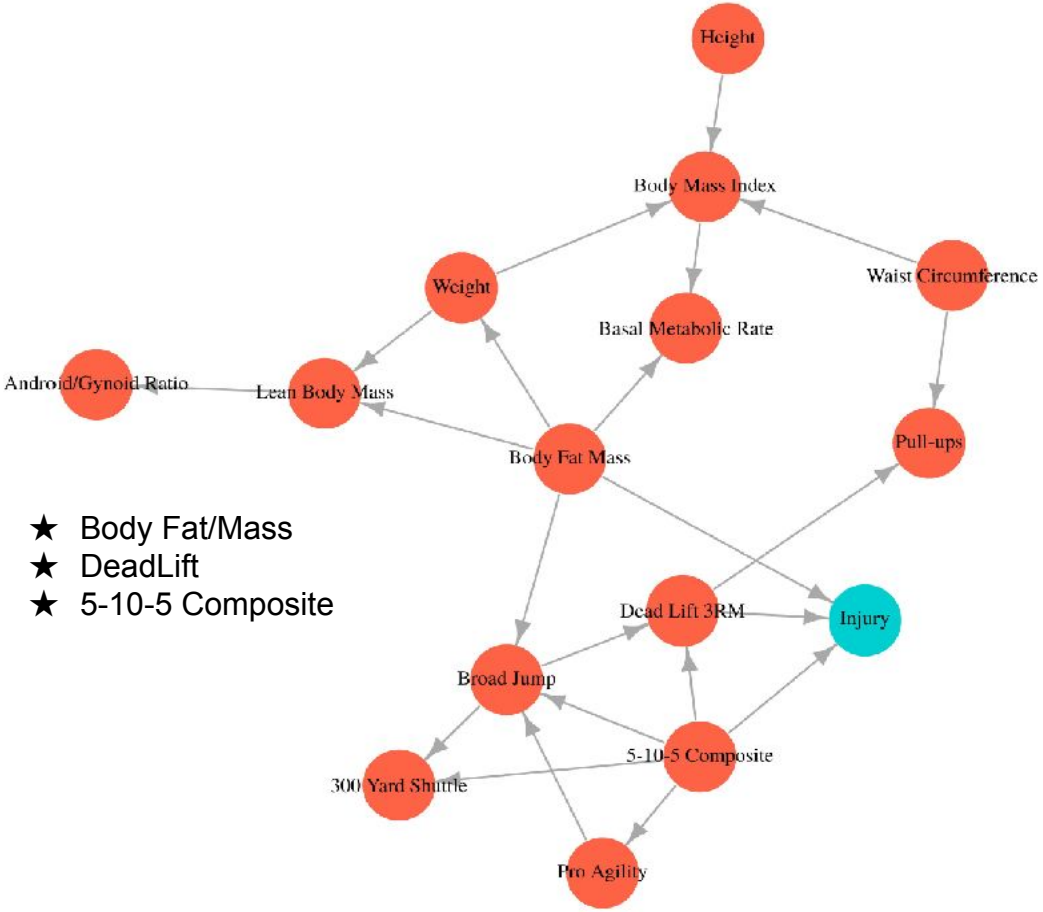
Warfighter: Annual / Point-in-time metrics

Prediction:

- Who
- When

Innate relationships:

- Why / how
- Dynamic prediction
- What-if simulation



★ Body Fat/Mass
★ DeadLift
★ 5-10-5 Composite

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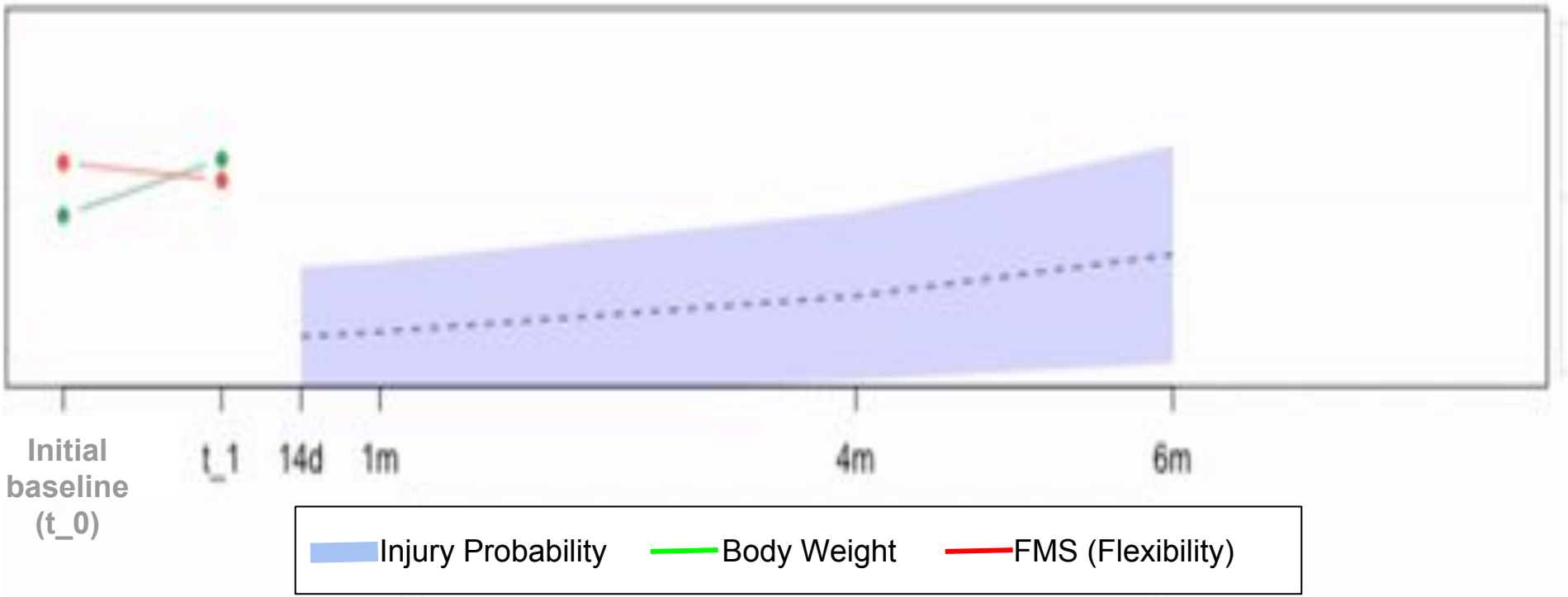
Bi-Variate Example: Dynamic Forecasts



Individualized dynamic and adaptive forecasts that continue to improve

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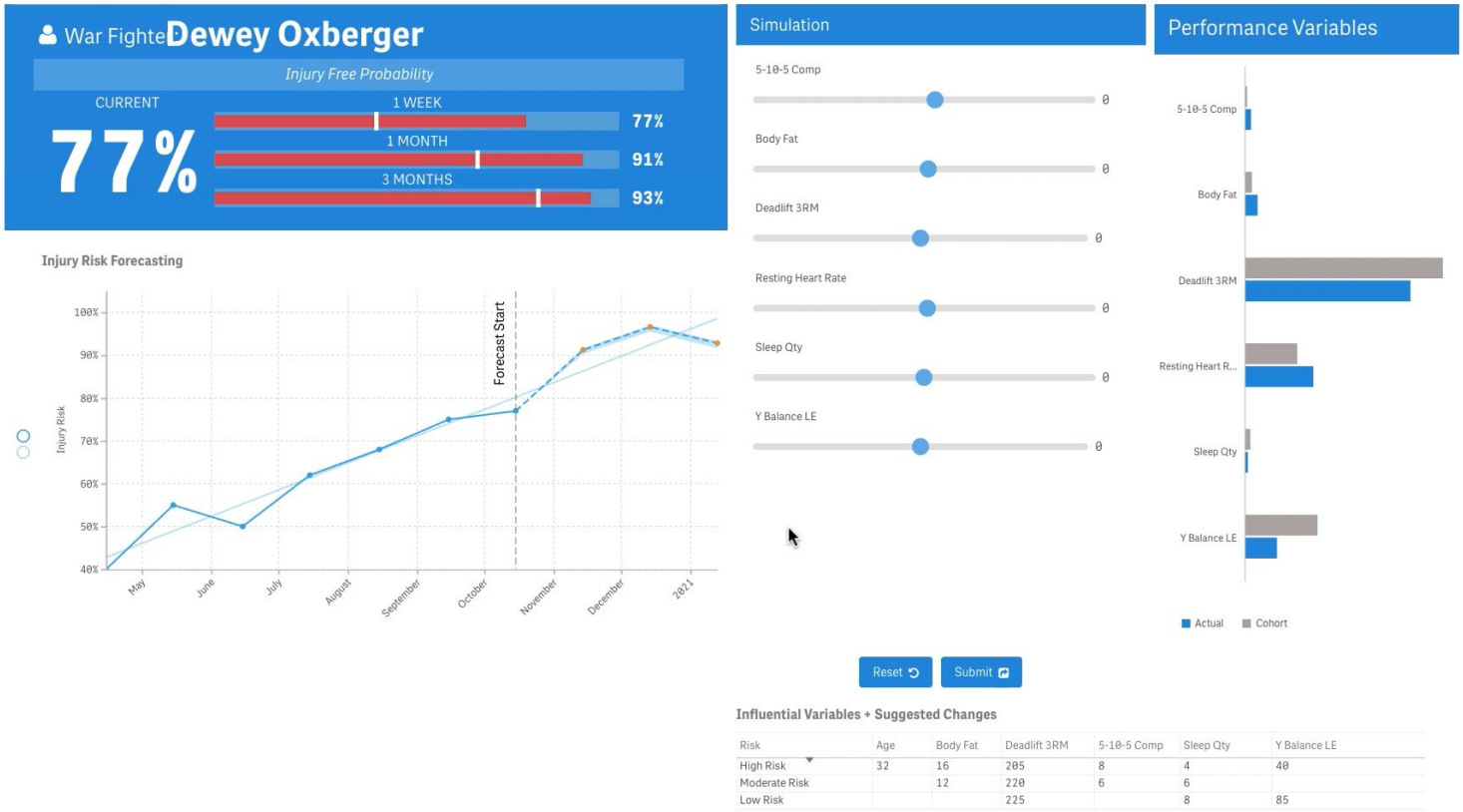
User Interface: Prototype



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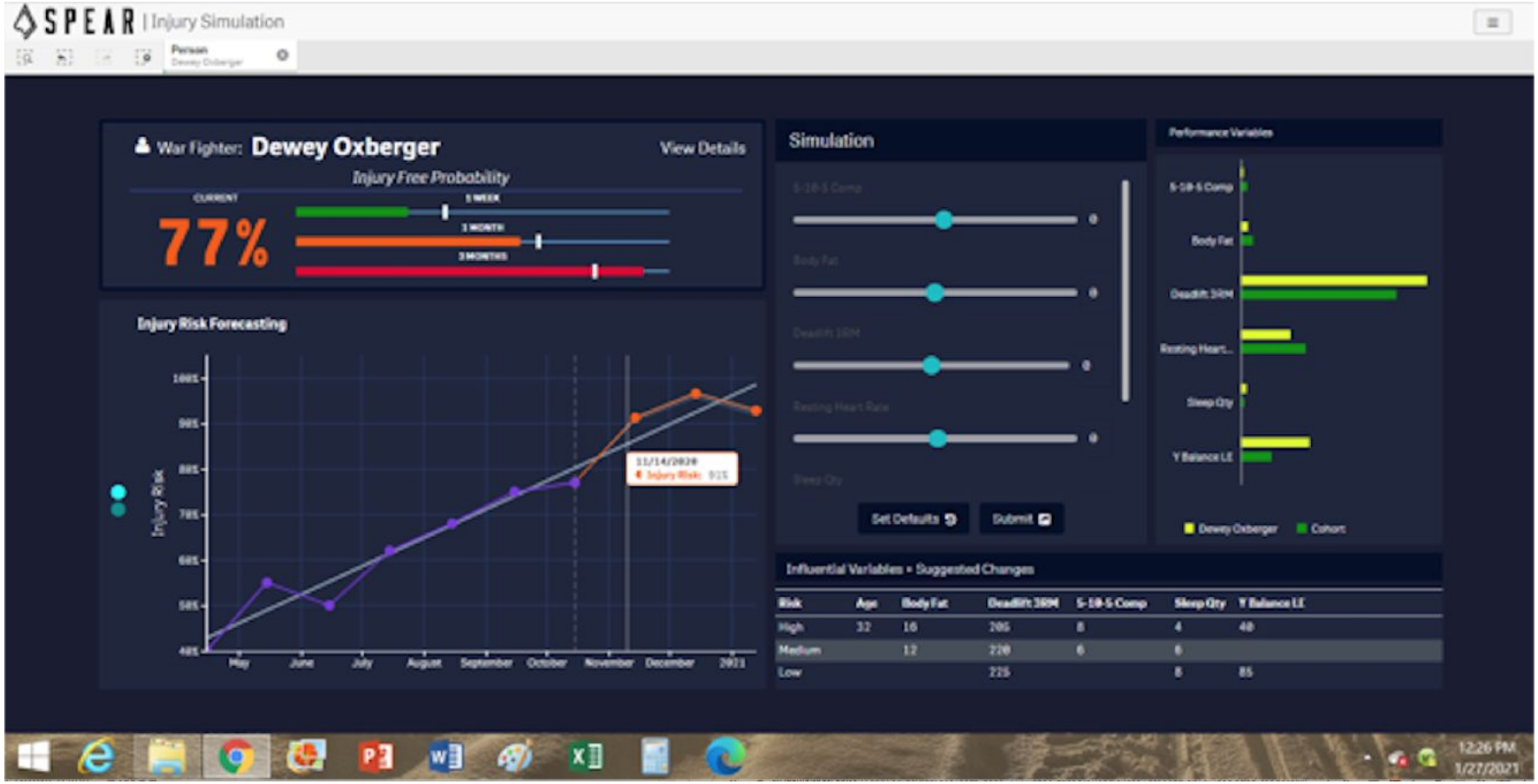
User Interface: Production



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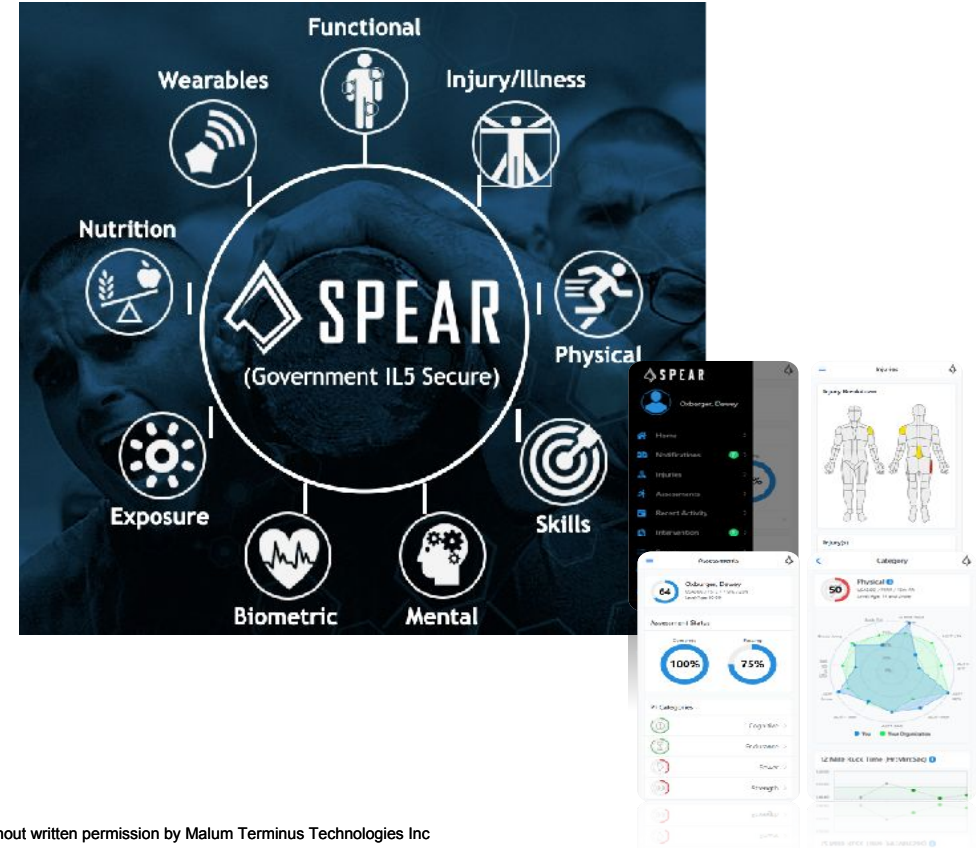
HP Tools: Providers, Leaders, Soldiers

Example: Keeping a high-potential BUD/S candidate training injury free

- Personnel readiness dashboard
- Risk level red (injury imminent)
- Stressor, sleep, weight loss & flexibility
- Simulation & intervention
- New protocols tested & implemented

Soldier **P**erformance **E**nhancement **a**nd **R**eadiness



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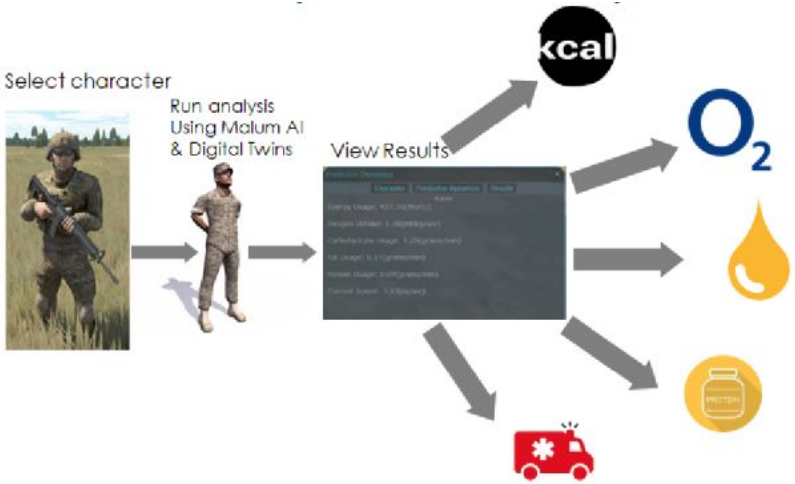
Discussion: Future applications, Q&A

- Readiness forecasting / planning
- Other ailments (e.g. PTSD, Self-harm)
- Performance optimization
- Annual assessments (e.g. ACFT, et al)
- Career tracts
- Mission planning
- Digital twins
- Others?

Mission Planning: AI-based risk reduction

- Forecasted Personnel Readiness
- Planners simulate various COA's
- Digital twins → optimize squads
- Waze for Mission Planning
- Improved human costs and outcomes



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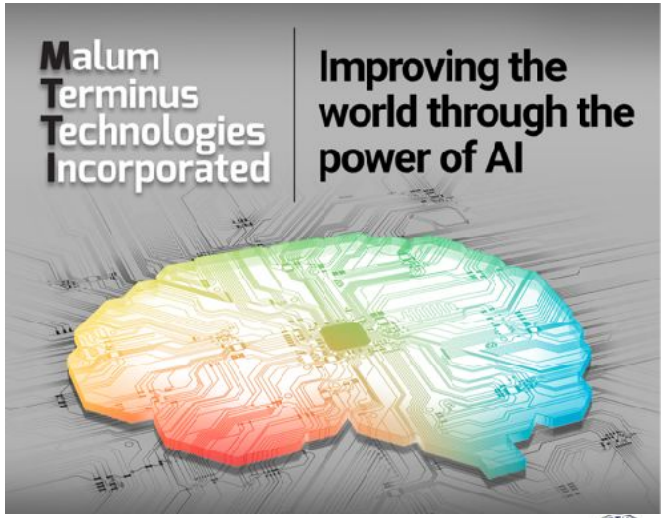
Thank You!



SPEAR

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